

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter presents the theoretical and empirical foundations of the study. It discusses the concepts of machine learning, supervised learning, predictive modelling, linear regression and multiple linear regression. It also reviews the mathematical formulation of regression using matrix algebra, least-squares estimation, categorical-variable encoding, numerical conditioning, standardization, sensitivity analysis and perturbation analysis.

The chapter further examines previous studies related to machine learning, regression and students' academic performance. The purpose is to establish the theoretical basis for applying matrix algebra to the prediction of students' examination scores and to identify the area addressed by the present study.

2.2 Concept of Machine Learning

Machine learning is a branch of computational science concerned with developing methods that enable computers to learn patterns and relationships from data and use the learned patterns to make predictions or decisions. Unlike traditional programming, where explicit instructions are provided for solving a problem, machine-learning methods allow a model to estimate relationships from available observations.

A machine-learning system generally consists of data, input variables, a target or output variable where applicable, a learning algorithm and a model obtained from the learning process. The model can subsequently be used to make

predictions on new observations.

Machine learning has applications in areas such as education, healthcare, finance, transportation, agriculture and engineering. In education, machine-learning techniques can be used to analyse factors associated with student performance and to predict academic outcomes.

The present study is concerned with the predictive application of machine learning. Student-related variables are supplied as input variables, while examination score is treated as the numerical output to be predicted.

2.3 Supervised Learning

Supervised learning is a type of machine learning in which an algorithm learns from observations for which the desired output is already known. The available observations consist of input variables and corresponding output values. The purpose of learning is to estimate a relationship between the inputs and output and then use this relationship to predict the output for new observations.

Mathematically, a supervised-learning problem can be represented generally as

$$y = f(X) + \varepsilon,$$

where X represents the input variables, y represents the observed output, f represents the underlying relationship between the inputs and output, and ε represents unexplained variation or error.

Supervised-learning problems are commonly divided into classification and regression. Classification deals with predicting categorical outcomes, such as whether a student passes or fails. Regression deals with predicting numerical

outcomes, such as examination scores.

Since the target variable in this study, exam_score, is numerical, the problem is formulated as a supervised-learning regression problem.

2.4 Concept of Matrix Algebra

Matrix algebra is a branch of mathematics concerned with operations involving matrices and vectors.

A matrix can be represented as

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

Matrices provide a compact method for representing systems involving several quantities.

Important operations include matrix addition, scalar multiplication, matrix multiplication and transposition.

The transpose of A is denoted by A^T .

Matrix multiplication is particularly important in regression because expressions such as

$$X^T X$$

And

$$X^T y$$

form the basis of the normal equations.

2.5 Matrices and Linear Systems

A system of linear equations can be represented in matrix form.

For example,

$$a_{11}x_1 + a_{12}x_2 = b_1$$

And

$$a_{21}x_1 + a_{22}x_2 = b_2$$

can be written as

$$Ax = b.$$

Matrix representation allows computational methods to be developed for solving systems containing many equations and unknowns.

The regression problem considered in this study is similarly represented as a matrix system.

2.6 Matrix Representation of Linear Regression

A simple linear regression model may be written as

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

For multiple explanatory variables, the model becomes

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \varepsilon_i.$$

The entire system can then be represented as

$$y = X\beta + \varepsilon.$$

This representation is particularly useful because the same mathematical operations can be applied regardless of the number of observations or predictors.

2.7 Multiple Linear Regression

Multiple linear regression models the relationship between a dependent variable and several explanatory variables.

The general form is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p + \varepsilon$$

The coefficient β_j represents the estimated change in the response associated with a one-unit change in x_j , conditional on the other variables in the model.

In matrix notation,

$$y = X\beta + \varepsilon$$

Multiple regression is useful where several factors may simultaneously contribute to the response variable.

2.8 Least-Squares Estimation

The least-squares method estimates the regression coefficients by minimizing the squared difference between observed and predicted values.

The residual vector is

$$e = y - X\beta$$

The least-squares problem is therefore

$$\min_{\beta} \|y - X\beta\|_2^2.$$

The solution provides the coefficient vector that gives the best fit under the least-squares criterion.

2.9 Normal Equations

Differentiating the least-squares objective leads to the normal equations

$$X^T X \hat{\beta} = X^T y.$$

If $X^T X$ is invertible, the solution is

$$\hat{\beta} = (X^T X)^{-1} X^T y.$$

The normal equations demonstrate the direct role of matrix algebra in regression.

However, forming $X^T X$ can increase numerical conditioning problems. This motivates the investigation of condition numbers and alternative computational methods.

2.10 Eigenvalues and Eigenvectors

For a square matrix A , an eigenvalue λ and corresponding eigenvector v satisfy

$$Av = \lambda v$$

Eigenvalues provide information about the structure of a matrix.

For the regression system, the eigenvalues of $X^T X$ are useful in examining the numerical scaling of the matrix.

The largest and smallest eigenvalues are particularly important when considering the condition number.

2.11 Matrix Conditioning and Condition Number

The condition number measures the sensitivity of a numerical problem to small changes in its input.

For a suitable matrix A , the condition number can be expressed as

$$\kappa(A) = \|A\| \|A^{-1}\|.$$

For a symmetric positive-definite matrix, the condition number in the 2-norm is related to its largest and smallest eigenvalues:

$$k(A) = \frac{\lambda_{\max}}{\lambda_{\min}}$$

A large condition number indicates that the numerical problem may be sensitive to perturbations.

2.12 Numerical Stability

Numerical stability concerns the extent to which computational results remain reliable when small numerical errors or perturbations occur.

In regression, poor conditioning can cause small changes in the design matrix or response vector to produce relatively large changes in estimated coefficients.

This makes numerical stability an important consideration when regression is formulated through matrix operations.

2.13 Standardization and Scaling

Variables may have substantially different numerical scales.

For example, age may be measured in years, attendance in percentages and study hours in hours per day.

Standardization transforms a variable using

$$z = \frac{x - \bar{x}}{s}$$

where $\bar{x}$ is the mean and s is the standard deviation.

Standardization can reduce differences in scale among variables and may substantially improve the conditioning of a matrix.

2.14 Perturbation Analysis

Perturbation analysis investigates how a mathematical solution changes when its input is slightly changed.

If a matrix X is replaced by

$$X + \Delta X,$$

the resulting coefficient vector may change.

Studying this change provides information about the stability of the regression solution.

In this study, perturbations were introduced into the explanatory variables and the resulting changes in coefficients and predictions were examined.

2.15 Academic Performance Prediction

Academic performance has been studied using numerous mathematical and computational techniques.

Student performance can depend on academic habits, attendance, study time, lifestyle factors and psychological characteristics.

Predictive modelling provides a means of examining how several variables are jointly related to an academic outcome.

However, the mathematical structure underlying these predictive models is also important, especially when the objective is to understand how computational results depend on the design matrix.

2.16 Review of Related Studies

Previous research on student academic performance has examined relationships involving study habits, attendance, lifestyle factors, mental health

and academic outcomes.

Many studies employ statistical or machine-learning methods to identify variables associated with academic performance.

However, the focus of the present study differs in that it places particular emphasis on the matrix formulation and numerical analysis of the regression model.

The study therefore combines predictive modelling with concepts from linear algebra and numerical analysis.

2.17 Empirical Review

Machine-learning techniques have increasingly been applied to educational data to predict student academic performance. Such studies commonly use information relating to students' demographic characteristics, academic activities, attendance, study behaviour and other factors as predictors.

Yağcı (2022) investigated the prediction of students' academic performance using educational data-mining and machine-learning techniques. The study demonstrated the usefulness of machine-learning methods for identifying patterns associated with academic outcomes.

Studies in the Nigerian educational context have also examined machine-learning approaches for predicting student performance. Research published in *Kybernetes* investigated the application of machine-learning techniques to student-performance prediction in higher educational institutions, demonstrating the growing relevance of predictive modelling in educational

decision-making.

More recent studies have continued to investigate regression and machine-learning approaches for academic prediction. Suleiman (2026), for example, examined the performance of linear regression in predicting academic outcomes, highlighting the continuing relevance of regression methods for educational prediction.

Other recent research has investigated regression-based data-mining approaches for student academic performance, while newer studies have explored explainable machine-learning models for educational prediction. These developments demonstrate that academic-performance prediction is an active area of research.

However, many predictive studies concentrate primarily on predictive accuracy and model comparison. The mathematical and numerical structure of the underlying regression computation may receive less attention. The present study therefore places particular emphasis on the matrix formulation of regression and the numerical properties associated with solving the resulting system.

2.18 Gap in Literature

Existing academic-performance studies commonly focus on predictive accuracy and identifying variables associated with student performance. Less attention is often given to the underlying matrix structure of the regression computation, particularly the relationship between

$$X, \quad X^T X,$$

their eigenvalues and their condition numbers.

This study addresses this gap by combining regression prediction with matrix-based numerical analysis.

In particular, it investigates:

- the construction of the design matrix;
- the least-squares solution;
- eigenvalue behaviour;
- condition numbers;
- standardization;
- perturbation effects; and
- prediction stability.

2.19 Summary of Literature Review

The literature reviewed in this chapter establishes the mathematical foundation for the present study.

Matrix algebra provides a natural representation of multiple linear regression, while least squares provides the mechanism for estimating the regression coefficients.

Eigenvalues and condition numbers provide tools for examining numerical properties, while standardization and perturbation analysis provide methods for investigating stability.

The concepts reviewed in this chapter provide the theoretical basis for the methodology and computational analysis presented in the following chapters.

CHAPTER THREE

METHODOLOGY

3.1 Introduction

This chapter presents the mathematical and computational methods employed in the study.

The study applies matrix algebra and multiple linear regression to model and predict students' examination scores from selected academic, behavioural, lifestyle and demographic variables.

Particular attention is given to construction of the design matrix, least-squares estimation, categorical-variable encoding, model evaluation, sensitivity analysis, perturbation analysis and numerical conditioning.

3.2 Research Design

The study adopts a quantitative computational research design.

The research involves the mathematical formulation of a multiple linear regression model and computational investigation of its numerical properties.

The study uses an existing dataset and applies mathematical and computational techniques to investigate the relationship between explanatory variables and examination scores.

3.3 Source of Data

The dataset used in this study was obtained from **Kaggle** and is titled Student Habits vs Academic Performance. The dataset was created by Jayanta Nath and contains student-related academic, behavioural, lifestyle and demographic variables.

The dataset contains 1,000 observations.

For model development and evaluation, the observations were divided into training and testing subsets, with 800 observations used for training and 200 observations used for testing.

Because the dataset is simulated, the results are interpreted as a demonstration of matrix algebra and regression methodology rather than as estimates of a real-world student population.

3.4 Variables Used in the Study

The dependent variable is:

Exam Score

The explanatory variable include:

Variable	Description
Age	Age of student
Study hours per day	Average daily study time
Social media hours	Average daily social-media usage
Netflix hours	Average daily Netflix/entertainment usage
Attendance percentage	Percentage of classes attended
Sleep hours	Average daily sleep
Exercise frequency	Frequency of exercise
Mental health rating	Mental-health rating
Gender	Gender Category
Part-time job	Whether student has a part-time job
Diet quality	Quality of diet
Parental education level	Parental education category
Internet quality	Quality of internet access
Extracurricular participation	Participation in extracurricular activities

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3.5 Data Preprocessing

The preprocessing procedure involved:

1. inspection of the dataset;
2. identification of numerical and categorical variables;
3. conversion of categorical variables to numerical dummy variables;
4. construction of the design matrix;
5. inclusion of the intercept column;
6. separation of explanatory and response variables; and
7. division into training and testing sets.

3.6 Encoding of Categorical Variables

Categorical variables were converted into numerical indicator variables.

For a categorical variable containing k categories, k-1 dummy variables were retained when an intercept was present.

For example, gender was represented using indicator variables for the Male and Other categories, while the remaining category served as the reference category.

The same procedure was applied to the other categorical variables.

3.7 Construction of the Design Matrix

The regression model was represented as

$$y = X\beta + \varepsilon$$

The design matrix had the form

$$X = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1,18} \\ 1 & x_{21} & x_{22} & \cdots & x_{2,18} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{800,1} & x_{800,2} & \cdots & x_{800,18} \end{bmatrix}.$$

The first column contains ones and represents the intercept.

The actual training design matrix had dimensions

800×19

Therefore

$$X \in \mathbb{R}^{800 \times 19}$$

The response vector was

$$y \in \mathbb{R}^{800}$$

Consequently,

$$X^T y \in \mathbb{R}^{19}$$

3.8 Formulation of the Regression Model

The multiple linear regression model can be written as

$$ExamScore = \beta_0 + \beta_1 Age + \beta_2 StudyHours + \beta_3 SocialMediaHours + \dots + \varepsilon$$

In matrix notation,

$$y = X\beta + \varepsilon$$

3.9 Least-Squares Estimation

The residual vector is

$$e = y - X\beta$$

The least-squares objective is

$$\min_{\beta} \|y - X\beta\|_2^2.$$

Differentiating the objective with respect to β and setting the result equal

to zero gives

$$X^T X \hat{\beta} = X^T y$$

If $X^T X$ is nonsingular,

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

3.10 Computation of the Coefficient Vector

The estimated coefficient vector contained 19 coefficients:

$\beta =$

6.997941
0.049258
9.531568
-2.681506
-2.318933
0.143439
1.976635
1.319862

1.953132
0.139663
0.807281
0.374590
-0.957974
-0.549902
-0.527839
-0.701798
-0.214797
-0.019558
-0.113722

3.11 Verification of the Least-Squares Solution

Three approaches were compared:

1. inverse-based solution;
2. direct solution of the linear system; and
3. Moore-Penrose pseudoinverse.

The differences between the solutions were of the order of 10^{-12} , indicating numerical agreement.

3.12 Prediction

Predictions were obtained using

$$\hat{y} = X\hat{\beta}$$

For an individual observation,

$$\hat{y} = x^T \hat{\beta}$$

3.13 Residual Analysis

Residuals were calculated using

$$e_i = y_i - \hat{y}_i$$

The residual vector was

$$e = y - \hat{y}$$

The mean residual was approximately

$$- 0.042841$$

3.14 Evaluation of Predictive Performance

The model was evaluated using MAE:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|,$$

MSE:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2,$$

RMSE:

$$RMSE = \sqrt{MSE},$$

and

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}.$$

3.15 Cross-Validation

Five-fold cross-validation was used to examine predictive stability.

The coefficient of determination was calculated for each fold and the mean and standard deviation were obtained.

3.16 Standardized Coefficient Analysis

Standardized coefficients were examined to compare predictor effects across variables measured on different scales.

3.17 Prediction Sensitivity Analysis

Sensitivity was examined by perturbing individual predictors and observing the resulting changes in predicted examination scores.

3.18 Perturbation Analysis

Small perturbations were introduced into the predictor variables. The resulting coefficient-vector changes were measured using the maximum absolute coefficient difference.

3.19 Eigenvalue Analysis

The eigenvalues of

$$X^T X$$

were computed.

The largest and smallest eigenvalues were used to examine the numerical structure of the matrix.

3.20 Condition Number Analysis

For the relevant matrices, the condition number was calculated.

For $X^T X$,

$$k = (X^T X) = \frac{\lambda_{max}}{\lambda_{min}}$$

3.21 Effect of Standardization on Conditioning

Numerical variables were standardized using

$$Z = \frac{x - \bar{x}}{s}$$

The condition numbers of the standardized matrices were then compared with the original values.

3.22 Numerical Stability of Regression Coefficients

Coefficient vectors obtained before and after relevant transformations were compared using:

$$\max_i \left| \hat{\beta}_i - \hat{\beta}_i' \right|$$

And

$$\left\| \hat{\beta} - \hat{\beta}' \right\|_2$$

3.23 Prediction Stability

Prediction stability was investigated by comparing predicted values before and after perturbation.

The maximum absolute prediction change and relative prediction change were calculated.

3.24 Summary of Computational Procedure

The overall procedure was:

Dataset → *Preprocessing* → *Design Matrix*

→ $X^T X$, $X^T y$ → $\hat{\beta}$ → $\hat{y}$

→ *Model Evaluation* → *Sensitivity* → *Perturbation*

→ *Eigenvalues* → *Condition Numbers* → *Standardization*

3.25 Software and Computational Tools

Python was used for the computational analysis.

The major libraries included NumPy, Pandas, Scikit-learn and Matplotlib.